



SIMATS
ENGINEERING



SIMATS
Saveetha Institute of Medical And Technical Sciences
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

CSA0618 – DESIGN AND ANALYSIS OF ALGORITHM

EXPERIMENT NO: 14

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Program:

```
def fast_power(x, n):  
    if n == 0:  
        return 1  
    half = fast_power(x, n // 2)  
    if n % 2 == 0:  
        return half * half  
    else:  
        return x * half * half  
  
x = int(input("Enter the base: "))  
n = int(input("Enter the exponent: "))  
result = fast_power(x, n)  
print("Result =", result)
```

Output:

The screenshot shows a code editor with a dark theme. The code is written in Python and is the same as the program shown above. The output window on the right shows the following text:

```
Enter the base: 2  
Enter the exponent: 10  
Result = 1024
```